APPLIED DATA SCIENCE CAPSTONE

Can the booster land again?

A data-led look at Falcon 9 first-stage recovery: from launch records and SQL to maps, dashboards, and classification.

DATA COLLECTION | EXPLORATION | PREDICTION

FALCON 9 FIRST-STAGE RECOVERY / FINAL PROJECT PRESENTATION



01 / EXECUTIVE SUMMARY

Reuse is measurable - but the evidence needs context.

Four takeaways, with each number tied to its source sample.

SQL LAUNCH RECORDS

101

Archived rows in the SQL analysis. Mission outcomes and booster landing outcomes are distinct measures.

MODELING SAMPLE

90

Rows and 83 encoded features in the classification notebook.

RECOVERY LABEL

60 / 90

Successful first-stage landings in the modeling sample: 66.7%.

REPORTED HOLDOUT

94.4%

17 of 18 predictions correct as reported; likely optimistic because tuning used all 90 rows.

Signal: recovery success rises sharply across the available years; the 90-row sample also shows higher success among later flight numbers.

Decision: use these results to prioritize better validation and operational questions - not as a flight-ready landing predictor.

The business question is about reusability.

Frame the outcome correctly before comparing models.

QUESTION

Which launch conditions are associated with a successful first-stage recovery?

Falcon 9's reusable first stage can reduce launch cost. For a launch provider evaluating a competitive bid, landing reliability matters because it changes how often hardware can be recovered and flown again.

TARGET: FIRST-STAGE LANDING

NOT: MISSION SUCCESS

DECISION PATH

1. **Collect**
   Combine API launch metadata with an archived launch-history table.
2. **Explain**
   Compare outcomes by time, site, orbit, payload, and landing context.
3. **Explore**
   Let reviewers filter the evidence geographically and interactively.
4. **Predict cautiously**
   Test whether launch attributes help classify recovery outcomes.

Interpretation guardrail: a successful mission can occur without a successful booster landing. SQL results count mission outcomes separately from the classification target.

Collection and wrangling turn launch history into a target.

Sources and transformations documented in the repository notebooks.

Data collection

A **SpaceX REST API**
Past launches plus related launch, rocket, payload, and landing-site metadata.

B **Archived launch table**
Wikipedia snapshot scraped for additional launch and landing-outcome fields.

C **Join on launch identifiers**
Retain Falcon 9 launches and fields needed for analysis.

Data preparation

+ Normalize date, site, orbit, booster, and payload fields.
+ Convert first-stage landing outcome to binary class: landed = 1, otherwise = 0.
+ Fill missing payload values with the sample mean in the provided workflow.
+ One-hot encode categorical orbit, site, landing-pad, and booster identifiers.
+ Result for modeling: 90 rows x 83 columns; scale features before classification.

DATA QUALITY
Missing payloads are imputed, not discarded.

TARGET ENGINEERING
Landing outcome is separated from mission outcome.

REPRODUCIBILITY
Notebooks preserve source queries, transforms, and plots.

EDA and interactive analytics triangulate the evidence.

Four methods answer different questions; none replaces the others.

SQL / SQLITE

QUERY

Group outcomes, payload mass, launch site, booster version, and mission results.

PANDAS / CHARTS

COMPARE

Plot success by flight number, payload, orbit, site, and year.

FOLIUM

LOCATE

Cluster launches at site coordinates and color markers by landing result.

PLOTLY DASH

FILTER

Choose launch site and payload range; inspect outcome and payload patterns.

Visual analytics

A site selector filters the success/failure view; a payload range slider narrows payload-versus-outcome points. Folium's marker clusters keep nearby site observations legible.

Predictive analysis

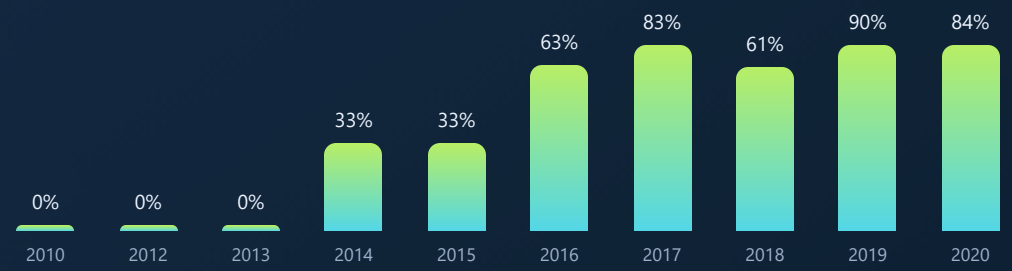
Train/test split, cross-validated hyperparameter search, and model scoring are recorded in the supplied classification notebook.

Evidence is drawn from separate saved snapshots; see the data-scope note on slide 12 before comparing counts.

Recovery improves with launch experience.

90-row modeling sample; rates are descriptive, not causal.

FIRST-STAGE SUCCESS RATE BY YEAR



Small early-year denominators (1-6 launches) make early rates volatile.

SUCCESS RATE BY LAUNCH SITE

CCAFS SLC 40		**33 / 55 60.0%**
KSC LC 39A		**17 / 22 77.3%**
VAFB SLC 4E		**10 / 13 76.9%**

Payload contrast: successful rows average 6,765 kg vs 4,785 kg for unsuccessful rows in this sample.

> Payload alone is not the explanation: site, orbit, booster generation, and time are entangled.

SQL separates mission performance from recovery.

Saved SQLite results from the archived launch table.

MISSION OUTCOMES

100 : 1

100 rows labeled mission success, 1 labeled mission failure in the query output.

NASA PAYLOAD

99,980 kg

Total payload mass for customers whose names begin with NASA.

CRS PAYLOAD

45,596 kg

Payload mass total for NASA Commercial Resupply Services missions.

Useful query cuts

+ Distinct sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.
+ Mean payload for F9 v1.1: about 2,535 kg.
+ First recorded ground-pad landing success in this table: 1 May 2017.

What SQL adds

It makes the unit of analysis explicit: a mission can succeed even when a landing attempt is absent or fails. Counting "mission success" is not a substitute for the recovery label.

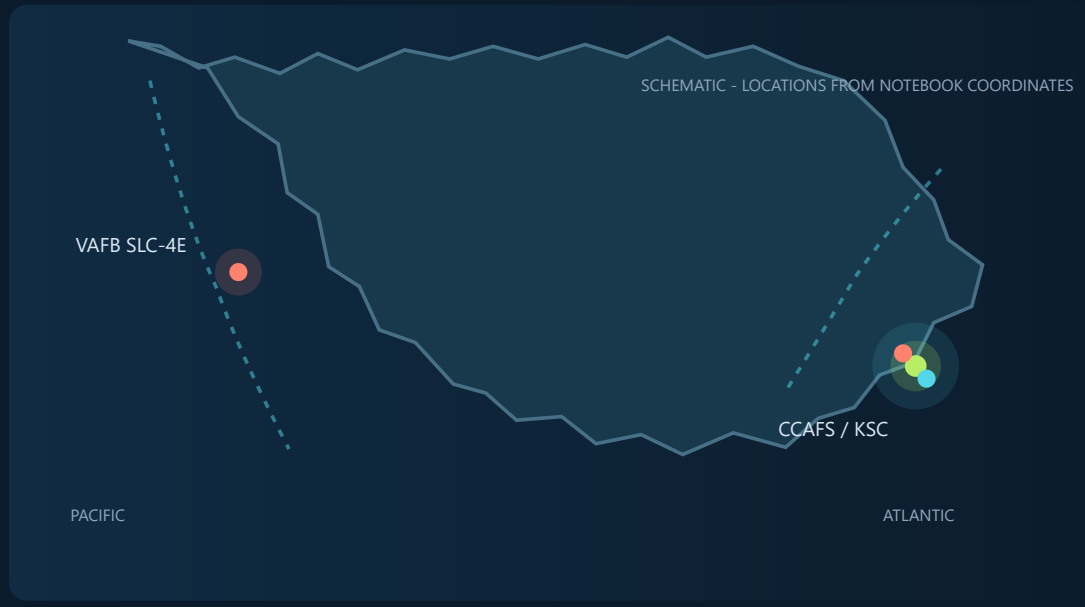
> **Interpretation:** retain both targets in reporting; do not describe the 100:1 mission count as a landing success rate.

SQL notebook: SELECT/GROUP BY and payload aggregates against SQLite table SPACEXTBL. Results are those saved in the cloned source notebooks.

07 / INTERACTIVE MAP RESULTS

Launch activity is geographically concentrated.

Folium site map: four launch-site coordinates, clustered outcome markers.



What the Folium view shows

Green markers indicate successful first-stage landings; red markers indicate unsuccessful outcomes. Marker clusters summarize launches around each site and can be explored at closer zoom.

SITES IN MAP

4

Florida dominates the map's launch density.

MAPPED SAMPLE

56

Rows in the saved geospatial/dashboard extract.

This is a schematic rendering for print. The repository notebook contains the interactive Folium map with marker clusters and coordinates.

Filters turn a static comparison into an exploration.

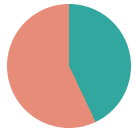
Dashboard extract contains 56 launch observations.

SpaceX Launch Records Dashboard

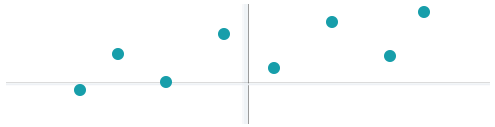
Launch site: All Sites v

Payload range: 0 - 10,000 kg

Success / failure by site



Payload mass vs. landing class



Site comparison in dashboard snapshot

CCAFS LC-40		7 / 26 26.9%
VAFB SLC-4E		4 / 10 40.0%
CCAFS SLC-40		3 / 7 42.9%
KSC LC-39A		10 / 13 76.9%

> The site pattern is a prompt for investigation, not a causal ranking: sample sizes differ, and this is not the same extract as the 90-row EDA/model sample.

Classification tests signal - the validation design matters.

Reported pipeline, with the key risk made explicit.

INPUTS

90 x 83

90 launches; 83 numeric/one-hot encoded features. Target: first-stage landing class.

SPLIT

72 / 18

80/20 train-test split with random_state = 2.

MODELS

LR / SVM

Logistic regression, support vector classifier, and decision-tree tuning.

Notebook workflow

1. **Scale features**
   StandardScaler transforms the encoded feature matrix.
2. **Search parameters**
   GridSearchCV uses 10-fold CV; examples include C for logistic regression and C/gamma/kernel for SVC.
3. **Score and inspect**
   Evaluate the selected estimator and plot a confusion matrix.

Validation risk

> **Leakage in saved workflow:** preprocessing and hyperparameter search are fit using all 90 observations before scoring the 18-row test split. The holdout is therefore not independent.

+ Put scaling inside a Pipeline.
+ Run GridSearchCV on training data only.
+ Use stratified folds and keep the test set untouched until final evaluation.
+ Report precision/recall and a confusion matrix, not accuracy alone.

Promising score; not yet a trustworthy forecast.

Results as recorded in the notebook, interpreted conservatively.

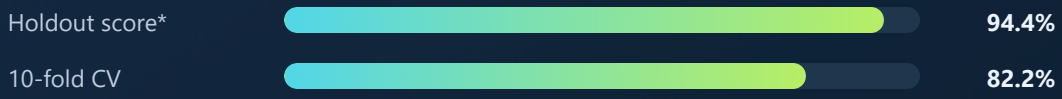
REPORTED TEST ACCURACY

94.4%

17 correct out of 18 test observations

Logistic regression and SVC both report this score; their saved 10-fold CV accuracy is about 82.2%.

REPORTED VS. EVALUATION CONFIDENCE



> ***Caveat:** The test metric is contaminated by full-data scaling/model tuning. It should not be presented as a reliable estimate of future-launch accuracy.

Next experiment: rerun the model with train-only preprocessing and tuning, then compare against a simple majority-class baseline.

Three snapshots, three questions - don't collapse them.

A data lineage check changes how every headline is read.

SQL SNAPSHOT

101

Archived rows used for mission/landing SQL queries.

MODEL SNAPSHOT

90

Encoded feature rows for recovery classification and EDA.

MAP/DASH SNAPSHOT

56

Rows in the saved geospatial and dashboard extract.

Why it matters

The snapshots differ in row counts and fields. A site rate from the 56-row dashboard should not be compared directly with a site rate from the 90-row model sample as if both came from one table.

Second signal: recovery is a learning curve

Year-level success rises from 0% in 2010-2013 to mostly above 80% in 2017-2020. Meanwhile success is higher for heavier payload rows in this sample - likely reflecting launch era, booster maturity, and mission mix rather than payload mass alone.

Novel takeaway: aligning snapshot provenance and separating mission outcome from recovery are prerequisites for a fair business comparison - and probably more valuable than adding model complexity.

Build the evidence chain before acting on the score.

What an analyst should take forward.

Conclusions

+ First-stage recovery becomes more common in later records, consistent with operational learning and hardware evolution.
+ Launch site, orbit, payload, booster, and time all shape the observed outcome; no single variable explains recovery.
+ SQL, visual EDA, Folium, and Dash make different parts of the story inspectable.
+ The 94.4% saved test score is not a dependable forecast because the pipeline leaks test information.

Recommended next steps

1. **Reconcile inputs**
   Publish one versioned analysis table and document row-level inclusion rules.
2. **Validate honestly**
   Use a leakage-safe pipeline and a time-aware holdout for future launches.
3. **Communicate uncertainty**
   Show denominators, class balance, calibration, and subgroup performance.
4. **Keep the tools live**
   Use the site/payload dashboard for exploration, not automated launch decisions.

Reproducible work, linked to the evidence.

Notebooks, scripts, data extracts, and the submitted deck.

PROJECT REPOSITORY

github.com/astrid1919/Data-Science-Capstone

Includes the completed capstone notebooks, Python dashboard, data extracts, and this presentation.

> Presentation: **SpaceX_Falcon9_Capstone_Presentation.pdf**

DATA AND METHODS REFERENCES

Course materials and completed notebook source:
github.com/lijqhs/data-science-notes

SpaceX API:
github.com/r-spacex/SpaceX-API

Launch history:
Archived Falcon 9 / Falcon Heavy launch table used in the capstone notebook.

All displayed metrics are reproduced from saved repository outputs or calculated directly from included CSV extracts; snapshots may differ in coverage.

Thank you.

Question for the next analysis: how much of the recovery learning curve remains when we evaluate only on future launches?